

722.6 TCU v2 — Shift-Class Architecture & SPC/MPC Control Specification

Purpose: Implementation spec for adding the four shift classes (power-on upshift, coast upshift, power-on downshift, coast downshift) to the existing ESP32 722.6 controller, with the SPC/MPC sequencing grounded in the ATSG 722.6 Technical Service Information manual (2004). Written for handoff to a code-assist environment working against the existing PlatformIO codebase (ShiftScheduler / SolenoidDriver / AdaptiveMemory / InputManager / SpeedReader / WebManager).

Design goal stated by the owner: maximum simplicity and sharp, confident shifts without sacrificing transmission health. The architecture below achieves "sharp" through correct fill + short inertia phases + freewheel-synchronized downshifts, NOT through brute pressure. Clutch thermal energy $\approx$ torque $\times$ slip-speed $\times$ slip-time; a *shorter*, correctly-filled shift is *gentler* on the clutch than a long soft one. Harshness comes from fill errors and overlap mistiming, not from short inertia phases.

0. Corrections to the previous review, per the ATSG manual

These override earlier review findings and **MUST** be reflected in the implementation:

1. **The Y4 "garage jiggle" is OEM-correct — keep it (refined).** ATSG p.53: *"3-4 Shift solenoid is pulsed continuously while in Park and during selector lever movement (Garage Shifts)." P.54 explains the mechanism: pulsing MV 3-4 pegs the B2 Shift Valve, routing fluid between B2's double pistons (counter-pressure) so apply force is reduced during N→D engagement. Once the lever rests in D, the solenoid shuts off and the B2 shift valve self-regulates counter-pressure down to ~0.5 bar for a firm hold. Refinement vs current code: pulse Y4 in **P** and during the **lever-movement window** (P/N-exit edge until ~800 ms after PRND settles in D or R), then stop. Do not pulse it forever in N at rest, and do not cut it the instant the lever leaves P/N (current code does both).*
2. **SPC resting state, corrected both ways.** ATSG solenoid chart (p.53): while driving in gear, SHIFT PC = **OFF** (de-energized). In P/N idle, SHIFT PC is pulsed at **≈33 % duty** and MOD PC at **≈40 % duty**. So: in-gear cruise → SPC de-energized (**setShiftPressure(100)** in this codebase's pressure-% API); P/N idle → SPC $\approx$ 67 % pressure-command ($\approx$ 33 % duty), MPC $\approx$ 60 % pressure-command ($\approx$ 40 % duty). This fixes the continuous-full-current heating issue *and* matches OEM garage-shift standby.
3. **20 ms pressure-update law.** ATSG p.80: *"The ETC can only change signal amplitude once per 20 milliseconds. This prevents overcorrection."* All SPC/MPC command

changes during shifts must be quantized to a 20 ms tick. The 1 kHz loop still samples sensors and evaluates exits every 1 ms, but pressure commands update only on 20 ms boundaries. This formalizes the fix for the near-instant "ramps."

4. **Turbine-speed constant is derivable from tooth counts (p.8).** n_2 = front carrier, n_3 = front sun (exciter rings per pp.32–33), turbine = front ring. Kinematics: $n_turbine = n_2 \cdot (1 + Z_s/Z_r) - n_3 \cdot (Z_s/Z_r)$. Small NAG (W5A330: sun 50T, ring 78T): **K = 1.6410**. Large NAG (58/92): K = 1.6304. Replace the current 1.61 with **1.641** and bench-verify TCC-locked (turbine vs engine within ± 20 RPM).
5. **Limp behavior nuance (p.91):** on an electrical fault the box *holds the current gear* until stop + ignition cycle, then gives 2nd + R. On a *mechanical/hydraulic* fault, 3rd only. Implication: after dropping all solenoids mid-drive, do **not** assert **current_gear = 2**; re-derive gear from live ratio (nearest-ratio classifier) once output > 200 RPM.
6. **1-2/4-5 solenoid is pulsed during ignition crank (p.53 note).** Add: pulse Y3 for ~300–500 ms at boot when engine RPM < 400.
7. **Adaptation framework should mirror OEM categories (p.77):** Accelerating upshifts, Deceleration upshifts, Accelerating downshifts, Deceleration downshifts — four separate adaptation spaces. Fill-time adaptation is stored in **20 ms cycles, max 15 cycles = 300 ms** (p.80); fill-pressure adaptation is a continuous trim (mbar analog, p.79). This class separation also structurally fixes the braking-decel "bind" pollution found in review (a braking coast downshift adapts only the coast-downshift table, with a metric appropriate to coast).
8. **Adaptation gating (p.78):** OEM relearn wants ATF 80–90 °C (60–105 °C acceptable), light throttle, level road. → Freeze all adaptive writes outside ATF 60–105 °C.

1. What SPC and MPC actually control (hydraulic model)

From ATSG pp.25–26, 52–58:

Solenoid	ATSG name	Type	No current →	Function
MPC (Y3/6y1)	Modulating Pressure Regulating Solenoid	Continuous VFS, PWM always	Max line pressure	Drives the Pressure Regulator Valve: line/working pressure 60→320 psi with torque.

Solenoid	ATSG name	Type	No current →	Function
				Also feeds the 1-2/4-5, 2-3 and 3-4 OVERLAP valves — i.e., MPC sets how firmly the <i>off-going</i> element holds/releases during overlap, "so that the shift overlap of a releasing and applying clutch corresponds to torque input" (p.52).
SPC (Y3/6y2)	Shift Pressure Regulating Solenoid	Continuous VFS, PWM during shifts only (driving)	Max shift pressure	Drives the Shift Pressure Regulator Valve: transitional clutch-apply pressure 0→220 psi for the oncoming element, routed only while a shift group is in its shift phase. Hydraulically blocked between shifts.
Y3/Y4/Y5	1-2/4-5, 3-4, 2-3 shift solenoids	On/off, normally closed	Blocked	ON = route shift-valve pressure to the

Solenoid	ATSG name	Type	No current →	Function
				command valve = initiate/perform shift. OFF in gear; command/holding valves latch the gear hydraulically.

Mental model for every shift: the shift solenoid opens the gate; **SPC is the right hand** (how hard the oncoming clutch is squeezed, phase by phase); **MPC is the left hand** (how much the off-going element resists letting go, plus overall line). Torque input scales both.

API convention used throughout this spec: the codebase's `setShiftPressure(pct)` / `setLinePressure(pct)` take *actual pressure %*; coil duty $\approx 100 - \text{pct}$. So "SPC 100 %" = de-energized = OEM "OFF" = also the max-apply lock command — conveniently, the end-of-shift seat command and the in-gear standby command are the same value.

Solenoid state reference table (OEM, p.53)

Condition	Y3 (1-2/4-5)	Y5 (2-3)	Y4 (3-4)	MPC	SPC
Ignition crank	pulsed	off	off	PWM	off
P/N idle	off	off	pulsed (Park + lever movement)	PWM ~40 % duty	PWM ~33 % duty
In gear, driving	off	off	off	PWM (line schedule)	OFF
During 1-2 / 4-5 (either direction)	ON for whole shift	off	off	PWM (overlap+line)	PWM (apply profile)

Condition	Y3 (1-2/4-5)	Y5 (2-3)	Y4 (3-4)	MPC	SPC
During 2-3 / 3-2	off	ON	off	PWM	PWM
During 3-4 / 4-3	off	off	ON	PWM	PWM

2. Torque estimation — the master input

All pressure maps key off **estimated input torque**, not raw TPS (OEM adaptation is denominated in Nm, p.77). With MAP (absolute, boost-capable) and RPM available:

$$T_est_nm = \text{clamp}(K_T * (\text{map_kpa} - \text{MAP_ZERO}), 0, T_MAX)$$

$$\text{load_pct} = 100 * T_est_nm / T_MAX$$

Starting calibration for the M111 2.3 Kompressor @ 1.2 bar (≈300–330 Nm peak, near the W5A330's design ceiling):

- **MAP_ZERO = 35 kPa** (closed-throttle manifold pressure)
- **K_T = 1.55 Nm/kPa** → ~295 Nm at 225 kPa absolute
- **T_MAX = 330 Nm**

Optional refinement later: small RPM correction table. Keep TPS for *intent* (kickdown detection, ROC), MAP for *torque*.

Class thresholds (with hysteresis):

- **POWER** if **tps > 8 %** AND **T_est > 25 Nm**
 - **COAST** if **tps < 5 %**
 - Between thresholds: keep previous class. Class is **latched at beginShift()** and never re-evaluated mid-shift.
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3. Shift classification

```
enum ShiftClass : uint8_t {  
  
    SC_POWER_UP,    // accelerating upshift (ATSG: up-shift under load)  
  
    SC_COAST_UP,    // lift-off upshift    (up-shift under no load)  
  
    SC_POWER_DOWN,  // kickdown / part-throttle downshift (under load)  
  
    SC_COAST_DOWN   // deceleration downshift (rolling to a stop / braking)  
  
};
```

Power-down sub-types (this is the key 722.6-specific insight):

From the clutch/sprag application chart (p.44) and power-flow summaries (pp.45–49): under *positive* torque, F2 carries the rear-sun reaction in 2nd and F1 carries the front-sun reaction in 1st; K3 (in 1st/2nd) and B1 (in 1st) are only required for engine braking.

- **PD_SPRAG** — **3→2 and 2→1 under power**: release the off-going clutch (K2 for 3-2, K1 for 2-1), let engine torque flare the turbine up, and the **freewheel catches at exactly synchronous speed** — drive is restored instantly with zero clutch slip through the speed change. The engine-braking element (K3 / B1) is then applied at near-zero slip differential. This is why a healthy 722.6 kickdown feels instant; exploit it. Sharp AND zero-wear.
- **PD_TIMED** — **4→3 (K3 off, B2 on) and 5→4 (B1 off, K1 on)**: clutch-to-clutch, no sprag assist. Requires controlled release-flare-catch with ratio feedback.
- Coast downshifts: sprags do NOT assist (torque is reversed in overrun) — but torque is tiny, so a gentle low-pressure timed apply is imperceptible.

Routing solenoid per group is unchanged from the existing

getRoutingSolenoidForShift() (Y3: 1↔2 and 4↔5; Y5: 2↔3; Y4: 3↔4). The solenoid stays ON for the entire shift, OFF at completion (hydraulic latch) — already correct in the codebase.

4. Phase sequences (the core of this spec)

All pressure commands quantized to the 20 ms tick. All `load`/torque bins captured at `beginShift()`. `t` = ms since shift start. Exit checks run at 1 kHz.

4.1 SC_POWER_UP — accelerating upshift

Phase	SPC (pressure %)	MPC (pressure %)	Duration / exit	Notes
PREP	hold standby	$\max(\text{cruise}, 40 + 0.5 \cdot \text{load})$	one 20 ms tick	Raise line/overlap authority <i>before</i> opening the gate. If load > 70 %: MPC = 100.
FILL	$\text{FILL_P}[\text{shift}] + \text{fill_p_trim}$ (high: 80–90)	hold	$\text{FILL_T}[\text{shift}] + \text{fill_t_trim}$ (base 120–180 ms; trim in ± 20 ms cycles, cap ± 100 ms)	Fire routing solenoid at phase entry. Stroke the piston, take up clearance. Ratio must NOT move yet.
TORQUE	$\text{APPLY} = 20 + 0.55 \cdot \text{load} + \text{apply_trim}$	hold	ratio departs old gear by 0.05, or 250 ms backstop	Oncoming takes over engine torque while off-going releases via overlap valve (MPC-governed) . Flare check: ratio rises > 0.10 above old gear → under-filled: command SPC +10

Phase	SPC (pressure %)	MPC (pressure %)	Duration / exit	Notes
				immediately, log flare , adapt fill_t_trim += 1 cycle.
INERTIA	ramp from APPLY: +2 %/tick low load → +4 %/tick high load; closed-loop ±1 %/tick on dRatio/dt vs target	hold	ratio ≤ target + 0.03	Target inertia duration: target_ms = clamp(400 - 1.5·load, 220, 400) . Optional torque-cut output asserted here (§9). Harshness check: inertia < 0.6·target OR output-decel spike → adapt apply_trim -= 2 .
LOCK	100 (= de-energized = max)	hold	120 ms	Seat the clutch.
END	stays 100 (standby)	decay to cruise schedule over 200 ms	—	Routing solenoid OFF (latch). Evaluate adaptation once, write dirty bit.

4.2 SC_COAST_UP — lift-off upshift

Same skeleton, gentler numbers: FILL at **FILL_P - 15** for **FILL_T - 20 ms**; TORQUE apply fixed at **25 %**; INERTIA ramp +1 %/tick, target 350 ms; MPC = cruise (no boost needed).

The oncoming clutch only has to pull the engine down; the risk is driveline clunk, not slip — keep pressures low. No flare adaptation from this class into the power tables.

4.3 SC_POWER_DOWN / PD_SPRAG — 3→2 and 2→1 under power

Phase	SPC	MPC	Exit	Notes
PREP	standby→ 10	$\max(\text{cruise}, 50 + 0.5 \cdot \text{load})$	1 tick	Force TCC open first if applied. Money-shift guard already passed in <code>beginShift()</code> .
RELEASE/FLARE	10 (oncoming unclamped)	hold	$\text{live_ratio} \geq \text{target} - 0.05$ (sprag catch) or 500 ms backstop	Fire routing solenoid. Off-going (K2/K1) exhausts via overlap valve; engine torque flares turbine; F2/F1 catches at sync — drive restored with zero commanded clamp. Detection: ratio reaches target and dRatio/dt collapses for ≥ 40 ms.
SECURE	30 → +2 %/tick → 100	hold	ratio within 0.05 of target for 100 ms	Apply the engine-braking element (K3 / B1) at near-zero

Phase	SPC	MPC	Exit	Notes
				slip. Gentle by construction.
END	100 standby	decay to cruise	—	Solenoid off, latch.

4.4 SC_POWER_DOWN / PD_TIMED — 4→3 and 5→4 under power

Phase	SPC	MPC	Exit	Notes
PREP	standby→ 20 (prefill, below clamp point)	$\max(\text{cruise}, 50 + 0.5 \cdot \text{load})$	1 tick	TCC open.
RELEASE/FLARE	20	hold	$\text{ratio} \geq \text{old} + 0.85 \cdot (\text{target} - \text{old})$, or 450 ms	Off-going releases, engine flares toward sync; oncoming sits prefilled but not clamping.
CATCH	$30 \rightarrow +3\%/\text{tick}$, closed-loop on dRatio/dt	hold	$\text{ratio} \leq/\geq \text{target} \pm 0.05$ held 60 ms	Clamp as sync approaches; the last ~15 % of the speed change is taken on the clutch.
LOCK/END	100 for 120 ms → standby	decay	—	Adapt: if ratio overshoot past target by > 0.15 (engine over-flared) → release_trim earlier clamp (start CATCH at 0.80 of ratio change); if catch

Phase	SPC	MPC	Exit	Notes
				shock detected → reduce CATCH ramp slope.

4.5 SC_COAST_DOWN — deceleration downshifts (incl. braking to a stop)

No sync wait — skip straight to a gentle catch. At closed throttle nothing flares the engine; waiting (the current DS_SYNC 400 ms) just delays every coast downshift.

Phase	SPC	MPC	Exit	Notes
PREP	standby→15	cruise	1 tick	
RELEASE	15	cruise	80 ms fixed (+ ds_timing_t rim)	
CATCH	15 → +1 %/tick	cruise	ratio at target ± 0.05, or 600 ms	Vehicle deceleration raises the ratio; the clutch drags the turbine up the rest of the way at minimal pressure. Loads are tiny — imperceptible by construction.
LOCK/END	100 / standby	cruise	—	Adaptation metric for this class must NOT be raw output-RPM drop (braking pollutes it). Use: output-decel

Phase	SPC	MPC	Exit	Notes
				during CATCH minus output-decel sampled in the 200 ms before the shift. Only the delta is the clutch's contribution.

Coast-down scheduler (auto downshifts while slowing): trigger by output-RPM thresholds so each catch lands at idle-friendly turbine speeds. Starting points (24-tooth output, RPM at the shaft): 5→4 below 1900, 4→3 below 1400, 3→2 below 900, 2→1 below 450 (or stay in 2 near stop to match the 2nd-gear-launch garage model — owner's choice; OEM "S" mode returns to 1st). Add 10 % hysteresis; suppress while a paddle request is pending; manual-mode ranges clamp the top gear but coast protection still forces downshifts at the floor thresholds.

4.6 Kickdown / forced power-down scheduling

TPS-ROC trigger (already implemented for line pressure) should also arm a kickdown evaluation: if $\text{tps} > 70\%$ and predicted turbine after a 1-gear (or 2-gear, executed sequentially) downshift ≤ 6000 RPM → request SC_POWER_DOWN. Multi-gear kickdowns execute as back-to-back single shifts (4→3 then 3→2), never skip-shifts — the valve groups are adjacent-gear only.

5. Starting calibration tables

// Fill (per upshift, pressure-% and ms) — K1, K2, K3, B1 volumes differ

FILL_P[4] = { 80, 82, 88, 78 }; // 1-2(K1), 2-3(K2), 3-4(K3 big drum), 4-5(B1)

FILL_T[4] = { 140, 150, 180, 130 };

// SPC torque-phase apply: $\text{APPLY\%} = 20 + 0.55 \cdot \text{load} + \text{apply_trim}$

```

// SPC inertia ramp:      slope%/20ms = 2 + 0.02*load  (i.e. 2..4)

// Inertia target:      ms = clamp(400 - 1.5*load, 220, 400)

// MPC cruise (line) — keep existing per-gear HOLDING_PRESSURE_MAP, but add:

//  during any shift: MPC = max(cruise, 40 + 0.5*load); load>70 → 100

//  after END: decay to cruise at 5%/20ms (no line-pressure thump)

// Standby duties (pressure-% API; duty ≈ 100 – pct):

//  driving in gear: SPC 100 (OFF), MPC per line schedule

//  P/N idle:      SPC 67 (~33% duty), MPC 60 (~40% duty)

//  crank (<400rpm): pulse Y3 for 400 ms at ~40% duty

```

ATSG fill-pressure adaptation ranges (p.79: 1-2 up to 1600 mbar vs 2-3 up to 800 mbar) imply the 1-2/K1 fill needs the widest pressure authority — give `fill_p_trim` a $\pm 15\%$ range on 1-2, $\pm 10\%$ elsewhere.

6. Adaptation v2

Replaces the current single up/down tables. Structure:

```
struct AdaptCell { int8_t fill_t_cycles; // ±5 (×20 ms, cap ±100 ms)
```

```
    int8_t fill_p_trim; // ±15 (%)
```

```
    int8_t apply_trim; }; // ±15 (%)
```

```
// Indexed [ShiftClass(4)][shift_idx(4)][torque_bin(4)]
```

```
// torque_bin from T_est at beginShift(): 0–25 / 25–50 / 50–75 / 75–100 % of T_MAX
```

Rules:

- **Bins captured at `beginShift()`** and carried to evaluation (fixes the completion-time misattribution bug).
 - One update per shift event; step sizes: ± 1 cycle, ± 2 %, ± 2 %. Deadband on detectors so clean shifts write nothing.
 - Two-sided convergence on power upshifts: flare $\rightarrow$ `fill_t +1` (primary) then `fill_p +2`; harsh (inertia too short / decel spike) $\rightarrow$ `apply -2`. No more one-way ratchet.
 - Coast-class detectors use the decel-delta metric (§4.5), never absolute output drop.
 - **Freeze all writes when ATF outside 60–105 °C** (p.78) or when in limp/failsafe.
 - Persist via the existing dirty-mask $\rightarrow$ NVS path, but flush on a 60 s timer or on entry to P/N, not per shift (NVS wear). Keep the atomic-mask fix from the review.
 - Web tuner: expose the new cells with per-field clamping on ingest.
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7. Garage / standby strategy (consolidated)

1. **In P:** pulse Y4 at ~ 37 % duty (current value is fine; ATSG doesn't publish the figure). SPC 67, MPC 60.
 2. **Lever movement window:** from the P/N-exit edge (existing PN-switch falling edge) until 800 ms after PRND settles in D or R: keep Y4 pulsing (B2 counter-pressure soft-apply per p.54), keep the 1500 ms engagement-grace slip mask.
 3. **Settled in D/R:** Y4 off (B2 shift valve regulates counter-pressure to ~ 0.5 bar $\rightarrow$ firm hold), SPC $\rightarrow$ 100 (OFF), MPC $\rightarrow$ cruise schedule.
 4. **In N at rest (not moving the lever):** Y4 off, SPC/MPC at P/N standby duties.
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8. Safety invariants (carried forward, with review fixes folded in)

- Money-shift guard on every downshift request: `output_rpm * target_ratio  $\leq$  6000`. Applies per single shift in multi-gear kickdowns.
- Overrev forced upshift > 6300 (gears 1–4); lug forced downshift; unchanged.
- Limp detection in **all forward ranges** (`isForwardRange()`, not just 'D'); on limp: all shift solenoids off, MPC 100 (de-energized = native max line), **SPC de-energized too** (`setShiftPressure(100)`, matching the all-off hydraulic failsafe), TCC off. Do not relabel gear to 2 — classify from live ratio; expect the box to hold the latched gear until stop (p.91).

- Reverse abuse: transition-based trigger only (R *entered* while output > threshold / drive previously engaged); legitimate-reverse latch otherwise (direction-blind sensor fix from review).
 - TCC forced open before any power downshift; TCC state frozen during all shifts; restore 300 ms after END.
 - ATF temperature gates: < 20 °C → FILL_T +40 ms, inertia targets +30 %, no TCC lock, adaptation frozen; > 120 °C → force shortest inertia targets, TCC locked where permitted, raise MPC floor +10.
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9. Optional but high-value: torque reduction via rusEFI

The OEM ETC requests engine torque reduction over CAN during the inertia phase of power upshifts. Since the engine runs rusEFI, wire one ESP32 GPIO → rusEFI digital input configured for shift retard (launch/ALS-style timing retard, e.g., -8° to -12°). Assert it for the INERTIA phase of SC_POWER_UP when load > 50 %. Effect: clutch absorbs meaningfully less energy per shift, allowing the short 220 ms WOT inertia target with *less* shock — the single biggest "sharp without sacrificing health" lever available in this build. (CAN-based torque requests are the cleaner long-term path given the wideband/CAN work already in progress.)

10. Implementation mapping to the existing codebase

- `TCU_Data.h`: add `ShiftClass`, `active_class`, `shift_load_bin/shift_rpm_bin` (captured at start), torque-estimate fields, new adaptation struct + telemetry fields (`t_est_nm`, `shift_class`, phase names).
- `ShiftScheduler`: replace the single PREFILL/OVERLAP/INERTIA and DS_* machines with one phase engine parameterized by class profile structs (per-phase: SPC entry value, ramp slope, MPC rule, exit predicate, backstop). Classification at `beginShift()`. Add the 20 ms quantizer (`if (ms_since_last_pressure_update >= 20)`), coast-down scheduler, kickdown arm.
- `SolenoidDriver`: add `setStandbyProfile(DRIVING | PARK_NEUTRAL)`, crank pulse for Y3, garage Y4 window control moved here.
- `AdaptiveMemory`: new 3-param cell layout, class-indexed; timed/parked flush; ingest clamps in `WebManager`.
- `SpeedReader`: $K = 1.641$; keep the bench-verification telemetry check.

- Prerequisite bug fixes from the prior review to land in the same pass: reverse-trigger transition logic, telemetry 10 Hz gate → 10 ms, `String` → fixed `char[]` + seqlock, dirty-mask atomics, `set_table` clamping, bin capture at shift start.
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11. Validation plan (bench → road)

1. **Bench, output free:** TCC-locked turbine formula check ($K = 1.641$); solenoid duty scope check of standby tables (P/N 33/40 %, in-gear SPC flat-off); 20 ms command quantization visible on SPC trace.
2. **Garage shifts:** N→D and N→R engagement < 800 ms, no clunk; verify Y4 window behavior on the scope; confirm B2 firmness after Y4 drop.
3. **Coast upshifts first** (lowest energy), then power upshifts at 25/50/75 % load — confirm flare detector silent, inertia durations near targets, adaptation cells move ≤ 2 steps then settle.
4. **Coast downshifts rolling to stop** — must be imperceptible; verify the decel-delta metric writes ~nothing during braking.
5. **Power downshifts:** 3→2 at part throttle — watch for the sprag catch signature (ratio snaps to target with SPC still at 10 %); only then enable 4→3/5→4 timed kickdowns; full kickdown last.
6. Log every shift: class, bins, per-phase durations, SPC/MPC traces, flare/harsh flags — the existing WebSocket dashboard at the corrected 100 Hz is sufficient.